

CONFUSION MATRIX:

Say Logistic Regression
Decision Tree etc.

EVALUATION METRIC FOR CLASSIFICATION

- HOW WELL A CLASSIFICATION MODEL IS PERFORMING
- Compares the predictions made by model where model was right or wrong.
- Helps in understanding where the model is making mistakes.

org. Out. Exp/Pred out.
↓ ↓
x₁ = MARKS y ŷ
52 + Pass + Pass TP
48 - Fail - Fail TN
59 + Pass - Fail FN
69 Pass Fail FNI
49 - Fail + Pass FP

Pass (+)
Fail (-)

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP) ✓	False Negative (FN) ✓ β = Type II Error	③ Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) ✓ α = Type I Error	True Negative (TN) ✓	④ Specificity $\frac{TN}{(TN + FP)}$
		① Precision $\frac{TP}{(TP + FP)}$	② Negative Predictive Value $\frac{TN}{(TN + FN)}$	⑤ Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

⑥ F₁-Score

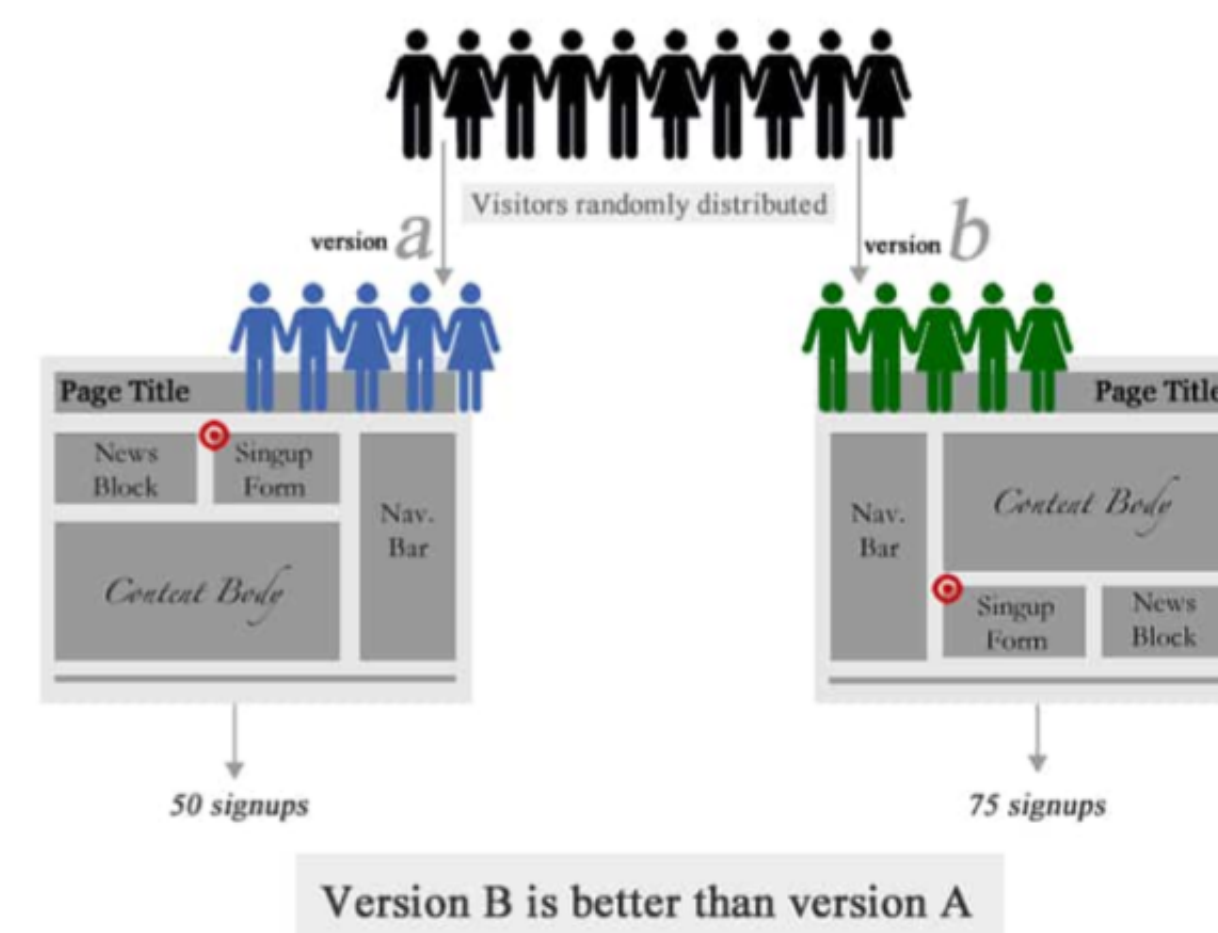
• Type-I Error (α)

• Type-II Error (β)

- **TP:** The model correctly predicts a positive class when it is indeed positive.
- **TN:** The model correctly predicts a negative class when it is indeed negative.
- **FP:** The model incorrectly predicts a positive class when it should have been negative.
- **FN:** The model incorrectly predicts a negative class when it should have been positive.

$$F_1\text{-Score} = 2 \left(\frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \right)$$

(Sensitivity = Recall)



Discussion

TASK:

• READ ABOUT FAMOUS "A/B TESTING"

CONFUSION MATRIX

EXAMPLE: Diabetes prediction ; logistic Regression ; Threshold ($p \leq 0.5 \rightarrow \text{Diabetes}(1)$; $p > 0.5 \rightarrow \text{Diabetes}(0)$)

Patient	Glucose X_1	BMI X_2	Actual Diabetes (y)	Predicted Probability $(p.p)$	Predicted Class $(\hat{y})$	
P1	160	32	1	0.88	1 $\rightarrow$ DIABETIC	TP
P2	145	30	1	0.75	1 $\rightarrow$ //	TP
P3	138	28	0	0.60	1 $\rightarrow$ //	FP
P4	120	25	0	0.40	0 $\rightarrow$ NOT DIABETIC	TN
P5	170	34	1	0.92	1	TP
P6	150	31	1	0.70	1	TP
P7	130	27	0	0.45	0	TN
P8	165	33	1	0.85	1	TP
P9	118	24	0	0.35	0	TN
P10	140	29	0	0.55	1	FP

SOLUTION:

- Identify TP, FP, TN, FN

TP=5
TN=3
FP=2
FN=0

TP = 5/10

- Make a Confusion Matrix

	PREDICTED DISEASE	PREDICTED HEALTHY
ACTUAL DISEASE	5	0
ACTUAL HEALTHY	2	3

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

- Calculation for Performance Measures {INTERPRETATION? ALSO}

1. SENSITIVITY: $\frac{5}{5+0} = 1$

2. SPECIFICITY: $\frac{3}{3+2} = \frac{3}{5}$

3. ACCURACY:

4. PRECISION:

5. F₁-SCORE:

Example No 2:
A sports analyst uses Logistic Regression to predict whether a cricket player will perform well (1) or not perform well (0) in an upcoming match based on:
{Practice Hours;Fitness Score}
The classification threshold is 0.5.

Player	Practice Hours	Fitness Score	Actual Performance	Predicted Probability
P1	5	80	1	0.85
P2	4	75	1	0.70
P3	3	65	0	0.55
P4	2	60	0	0.40
P5	6	85	1	0.90
P6	5	78	1	0.75
P7	3	68	0	0.45
P8	6	88	1	0.92
P9	2	58	0	0.35

- 1) Convert the predicted probabilities into predicted classes using threshold = 0.5.
- 2) Construct the Confusion Matrix showing:
- 3) Calculate the following performance measures:

Accuracy;Precision;Recall (Sensitivity)
- 4) Interpret the results:

• Is the model better at identifying good performers or poor performers?

• Which type of error is more significant in sports analytics? $(\alpha \text{ or } \beta)$